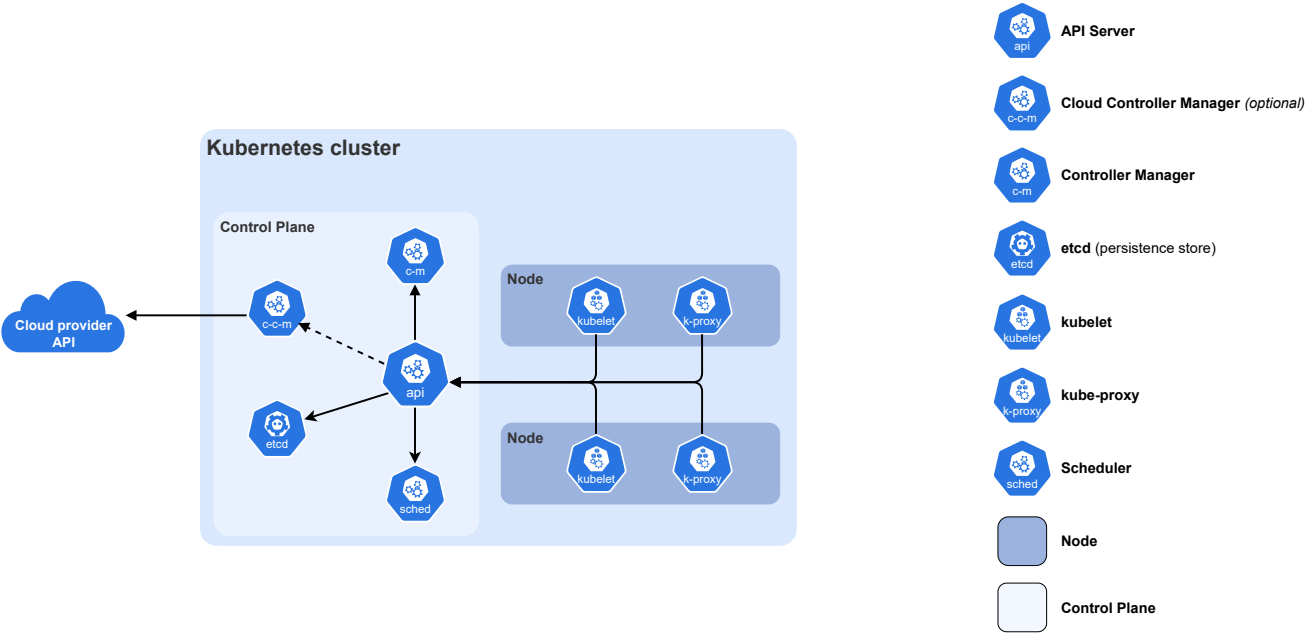




Kubernetes Components

An overview of the key components that make up a Kubernetes cluster.

This page provides a high-level overview of the essential components that make up a Kubernetes cluster.



The components of a Kubernetes cluster

Core Components

A Kubernetes cluster consists of a control plane and one or more worker nodes. Here's a brief overview of the main components:

Control Plane Components

Manage the overall state of the cluster:

kube-apiserver

The core component server that exposes the Kubernetes HTTP API.

etcd

Consistent and highly-available key value store for all API server data.

kube-scheduler

Looks for Pods not yet bound to a node, and assigns each Pod to a suitable node.

kube-controller-manager

Runs controllers to implement Kubernetes API behavior.

cloud-controller-manager (optional)

Integrates with underlying cloud provider(s).

Node Components

Run on every node, maintaining running pods and providing the Kubernetes runtime environment:

kubelet

Ensures that Pods are running, including their containers.

kube-proxy (optional)

Maintains network rules on nodes to implement Services.

Container runtime

Software responsible for running containers. Read [Container Runtimes](#) to learn more.



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Your cluster may require additional software on each node; for example, you might also run [systemd](#) on a Linux node to supervise local components.

Addons

Addons extend the functionality of Kubernetes. A few important examples include:

DNS

For cluster-wide DNS resolution.

Web UI (Dashboard)

For cluster management via a web interface.

Container Resource Monitoring

For collecting and storing container metrics.

Cluster-level Logging

For saving container logs to a central log store.

Flexibility in Architecture

Kubernetes allows for flexibility in how these components are deployed and managed. The architecture can be adapted to various needs, from small development environments to large-scale production deployments.

For more detailed information about each component and various ways to configure your cluster architecture, see the [Cluster Architecture](#) page.

Feedback

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No

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